RAHUL NIVASH

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EDUCATION

University of Manitoba **Winnipeg, MB**
B.Sc. (Honours) Joint Computer Science & Physics, Minor in Mathematics *Expected Dec 2026*

- **Academic Standing:** Institutional GPA: 3.55 / 4.50; Dean's Honour List (Winter 2025: 4.00 / 4.50).
- **Selected Coursework:**
  - *Optics, Electronics & Applied Physics:* Circuit Theory & Electronics (A+), Optics (A), Waves & Modern Physics (A+), Electro- and Magnetostatics (A), Classical Mechanics (A), Mathematical Physics I & II (A, A), Medical Physics & Physiological Measurement (A), Quantum Mechanics II & III (B+, B+).
  - *Computation & Systems:* Computational Physics (A), Computer Organization (A), Data Structures & Algorithms (A), Computer Graphics (A).
  - *Mathematics:* Multivariable Calculus (A), Vector Geometry & Linear Algebra (A), Undergraduate Research in Mathematics (A).
- **Fall 2026 (In Progress):** Advanced Algorithms, Statistical Mechanics, Cryptography and Cryptosystems, Computer Graphics 2.

RESEARCH EXPERIENCE

Undergraduate Research Fellow (Faculty of Science USRA) **Summer 2026**
Department of Mathematics, University of Manitoba *Winnipeg, MB*
Supervised by Dr. Christopher Heggerud

- Engineered a GPU-accelerated numerical solver in Python (`JAX`) on HPC clusters to compute trajectories for high-dimensional nonlinear dynamical systems.
- Computed full Lyapunov exponent spectra via QR decomposition to classify stability regimes and their transitions (e.g., chaos, limit cycles, and steady states).
- Sampled a high-dimensional parameter space using Latin Hypercube Sampling (LHS) to resolve regime boundaries.
- Trained a gradient boosting machine learning surrogate model (`XGBoost`) with active learning to detect transition boundaries, cutting numerical simulation runtime by orders of magnitude.

Undergraduate Researcher – Nonlinear Optics & HPC (Independent Project) **Jan 2026 – Jun 2026**
Department of Physics & Astronomy, University of Manitoba *Winnipeg, MB*

- Extended computational physics (`PHYS 4250`) coursework modeling optical turbulence in nonlinear ring cavities into an independent project focused on high-performance numerical methods.
- Formulated a 2D real-valued system to eliminate complex-arithmetic overhead, migrating the solver to C++ and `CUDA` to parallelize large-scale parameter sweeps.
- Derived Jacobian matrices analytically to compute Lyapunov exponents and determine stability thresholds, validating results against continuous-wave theory.
- Implemented a box-counting algorithm to compute the attractor's fractal dimension and validated numerical convergence against universal scaling laws.

Undergraduate Researcher (MATH 4920) **Winter 2026**
Manitoba eXperimental Mathematics Laboratory (MXML), University of Manitoba *Winnipeg, MB*
Supervised by Dr. Christopher Heggerud

- Modeled nonlinear class-B laser cavity equations and demonstrated their dynamical equivalence to a coupled 3-variable tritrophic system exhibiting transient chaos.
- Implemented high-order adaptive Runge-Kutta integration in Python (`SciPy`) to accurately capture sensitive strange attractors.
- Evaluated localized Finite-Time Lyapunov Exponents (FTLEs) to quantify predictability horizons and modeled state transitions using Markov chains.
- Constructed Poincaré maps and calculated attractor fractal dimensionality using spatial neighbor search algorithms, defending findings in an oral presentation to department faculty.

PUBLICATIONS & PREPRINTS

Nivash, R., & Heggerud, C. *Machine Learning Surrogate Modeling for Mapping Transient and Asymptotic Regimes in High-Dimensional Dynamical Systems.* Manuscript in preparation (Expected submission Dec 2026).

CONFERENCE PRESENTATIONS & TALKS

Theory Canada 18 **Jun 19–20, 2026**
Nivash, R. *Université de Montréal, Montreal, QC*
Oral Presentation (Contributed Talk)

High-Performance Computational Modeling of Optical Turbulence and Discrete-Time Chaos in the Ikeda Map

Canadian Undergraduate Physics Conference (CUPC) **Oct 15–19, 2026**
Nivash, R. *McMaster University, Hamilton, ON*
Poster Presentation
Optical Turbulence and Chaos in the Ikeda Map: High-Performance Computational Modeling of Discrete-Time Chaos

Maud Menten Institute (MMI) HQP Summit **Aug 14–16, 2026**
Nivash, R., & Heggerud, C. *University of Victoria, Victoria, BC*
Poster Presentation (Funded Presenter)
Studying Regime Shifts in Nonlinear Ecosystems: Mapping Topological Boundaries in a 9D Hyperspace Explicitly and with Machine Learning

Undergraduate Research Showcase **Oct 22, 2026**
Nivash, R., & Heggerud, C. *University of Manitoba, Winnipeg, MB*
Poster Presentation
Studying Regime Shifts in Nonlinear Ecosystems: Mapping Topological Boundaries in a 9D Hyperspace Explicitly and with Machine Learning

Math Biology Research Group Meeting **Sep 14, 2026**
Nivash, R., & Heggerud, C. *University of Manitoba, Winnipeg, MB*
Oral Presentation
Regime Shifts in Non-Linear Ecosystems: Topological Mapping with Machine Learning

Summer Undergraduate Research Experience (SURE) Poster Session **Aug 19, 2026**
Nivash, R., & Heggerud, C. *University of Manitoba, Winnipeg, MB*
Poster Presentation
Studying Regime Shifts in Nonlinear Ecosystems: Mapping Topological Boundaries in a 9D Hyperspace Explicitly and with Machine Learning

TECHNICAL SKILLS

HPC & Hardware Acceleration:	Designing GPU-accelerated simulation pipelines, parallel numerical solvers, and vectorized parameter sweeps on Linux compute clusters.
Optics & Physical Modeling:	Modeling nonlinear optical ring cavities, class-B laser rate equations, and wave/stability dynamics; foundational training in circuit analysis and electronics.
Scientific Machine Learning:	Developing physics-informed surrogate models (Gradient Boosting, MLPs, Gaussian Processes) and active-learning pipelines for high-dimensional optimization.
Numerical & Stability Methods:	Implementing high-order adaptive ODE solvers, analytical Jacobian differentiation, Lyapunov spectrum algorithms, and Latin Hypercube Sampling.
Software, Languages & Tools:	C/C++, `CUDA`, Python (`JAX, NumPy, SciPy, XGBoost`), Java, Linux, `Git`, LaTeX.

TEACHING, LEADERSHIP & COMMUNITY SERVICE

Founder & Lead Instructor **Jan 2026 – Present**
SpeakUp! Learning *Winnipeg, MB (Hybrid)*

- Founded an international educational consultancy focusing on developing conversational English skills.
- Directing a growing team of instructors to deliver customized curricula to a recurring international client roster.
- Designed an intensive 2-week fluency framework from first principles, managing operational logistics, instructional delivery, and client relations.

Private STEM Tutor **Nov 2025 – Present**
TutorBright *Winnipeg, MB*

- Deliver tailored one-on-one instruction across university-level mathematics, physics, and chemistry, emphasizing foundational conceptual logic.

Independent Academic Consultant – Mathematics **Mar 2026 – May 2026**
Independent Practice *Winnipeg, MB*

- Coached students in advanced competitive mathematics, directing targeted preparation for the University of Waterloo Gauss Mathematics Competition.
- Developed instructional frameworks covering discrete combinatorics, probability distributions, and algorithmic problem-solving logic.

Ski Patroller (Volunteer) **Sep 2025 – Apr 2026**
Canadian Ski Patrol (Red River Zone) *Winnipeg, MB*

- Certified in Advanced First Aid and CPR (BLS); executed on-hill emergency response, accident triage, and safety risk mitigation under high-pressure conditions.